

Non-NCBI-Databases

Name	Type	Description
EMBL / ENA	Primary	The European Nucleotide Archive, comprising the EMBL Nucleotide Sequence Database and the Ensemble Trace Archive, is a collection of DNA sequences https://www.ebi.ac.uk/ena/browser/home
DDBJ	Primary	Stands for DNA Data Bank of Japan and collects DNA sequences. It is located at the National Institute of Genetics (NIG) in the Shizuoka prefecture of Japan https://www.ddbj.nig.ac.jp/index-e.html
GOLD	Primary	Genomes On Line Database is a comprehensive resource for centralized monitoring of genome and metagenome projects worldwide https://gold.jgi.doe.gov/
UniProt	Primary	A comprehensive resource for protein sequence and annotation data. Collaboration between the European Bioinformatics Institute (EMBL-EBI), the SIB Swiss Institute of Bioinformatics and the Protein Information Resource (PIR). https://www.uniprot.org/
RCSB-PDB	Primary	The US data center for the global Protein Data Bank (PDB) archive of 3D structure data for large biological molecules (proteins, DNA, and RNA) https://www.rcsb.org/

PDBe	Primary	The european data center for the global Protein Data Bank (PDB) archive of 3D structure data for large biological molecules (proteins, DNA, and RNA) https://www.ebi.ac.uk/pdbe/
PDBj	Primary	The japanese data center for the global Protein Data Bank (PDB) archive of 3D structure data for large biological molecules (proteins, DNA, and RNA) https://pdj.org/
BMRB	Primary	Collects NMR data from any experiment and captures assigned chemical shifts, coupling constants, and peak lists for a variety of macromolecules https://bmr.io/
EMDB	Primary	Collects 3D volumes & associated information of macromolecular complexes & subcellular structures from electron cryo microscopy & electron cryo tomography https://www.ebi.ac.uk/emdb/
STRING	Secondary	Database of known and predicted protein-protein interactions. The interactions include direct (physical) and indirect (functional) associations https://string-db.org/
InterPro	Secondary	Provides functional analysis of proteins by classifying them into families and predicting domains and important sites https://www.ebi.ac.uk/interpro/
OMIM	Secondary	Online Mendelian Inheritance in Man (OMIM®) is a continuously updated catalog of human genes and genetic disorders and traits https://www.omim.org

Gene Ontology	Secondary	World's largest source of information on the functions of genes https://geneontology.org
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NCBI-Databases

Name	Type	Description
Protein	Primary	This database contains protein sequences derived from several sources https://www.ncbi.nlm.nih.gov/protein/
Gene	Primary	This database contains information about many different genes in many different species https://www.ncbi.nlm.nih.gov/gene/
PubMed		A free resource supporting the search and retrieval of biomedical and life sciences literature with the aim of improving health https://pubmed.ncbi.nlm.nih.gov/
PMC		A free full-text archive of biomedical and life sciences journal literature at the U.S. National Institutes of Health's National Library of Medicine (NIH/NLM) https://www.ncbi.nlm.nih.gov/pmc/
Genome		Contains sequence and map data from the whole genomes of over 1000 species or strains https://www.ncbi.nlm.nih.gov/genome/
Nucleotide	Primary	This database is a collection of nucleotide sequences (RNA / DNA) stemming from a multitude of different sources https://www.ncbi.nlm.nih.gov/nucleotide/
Genbank		GenBank® is the NIH genetic sequence database, an annotated collection of all publicly available DNA sequences https://www.ncbi.nlm.nih.gov/genbank/

Other Databases (not) covered in the course

Name	Type	Description
Ensembl	Secondary	Ensembl is a genome browser for vertebrate genomes that supports research in comparative genomics, evolution, sequence variation and transcriptional regulation https://www.ensembl.org/index.html
UCSC	Secondary	The UCSC Genome Browser is a web-based tool serving as a multi-powered microscope that allows researchers to view all 23 chromosomes of the human genome at any scale from a full chromosome down to an individual nucleotide. https://genome.ucsc.edu/index.html
AlphaFoldDB	Primary	AlphaFold DB provides open access to over 200 million protein structure predictions to accelerate scientific research https://alphafold.ebi.ac.uk/
ModelArchive	Primary	Since 2006, only structures that have been determined experimentally are allowed to be deposited in the PDB, and theoretical models of macromolecular structures are no longer part of the PDB archive. ModelArchive is being developed following a community recommendation during a workshop on applications of protein models in biomedical research. https://www.modelarchive.org/
PHI-Base	Secondary	curated molecular and biological information on genes proven to affect the outcome of pathogen-host interactions http://www.phi-base.org/index.jsp

Ensembl Genomes	Secondary	Providing genome data for non-vertebrate species, with tools for the manipulation, analysis and visualisation of that data (Plants, Metazoa, Protists, Fungi & Bacteria) http://ensemblgenomes.org
BioGRID	Secondary	public database that archives and disseminates genetic and protein interaction data from model organisms and humans https://wiki.thebiogrid.org/doku.php/aboutus
IMP	Secondary	Analyze your experimental results in the functional context of gene-gene networks from multiple organisms http://imp.princeton.edu
SCOP	Secondary	The SCOP database aims to provide a detailed and comprehensive description of the structural and evolutionary relationships between all proteins whose structure is known https://scop.mrc-lmb.cam.ac.uk/
SMART	Secondary	Simple Modular Architecture Research Tool allows the identification and annotation of protein domains http://smart.embl-heidelberg.de/
ProSite	Secondary	A database of protein families and domains that is based on the observation that, while there is a huge number of different proteins, most of them can be grouped, on the basis of similarities in their sequences, into a limited number of families. https://prosite.expasy.org/
eggNOG	Secondary	A public database of orthology relationships, gene evolutionary histories and functional annotations http://eggnog5.embl.de/#/app/home